These are the actual **enterprise-grade engines** use in **production-grade, real banking/financial messaging clients** used for **server-to-server**, **host-to-host**, or **direct IP-to-IP** delivery of ISO 20022, MT103, and other financial payloads.

✅ 1. SWIFT Production Messaging Engines (Bank-to-Bank)

These are the **most authoritative** financial messaging clients in the world.

A. SWIFT Alliance Access (SAA)

- Primary production SWIFT FIN/MX gateway.
- Handles MT103, MT202, MX pacs.008, pacs.009, camt.*.
- Integrates with HSMs for signatures (HSM-SA).
- High throughput, clustering, HA.

B. SWIFT Alliance Gateway (SAG)

- Connects internal bank systems to SWIFTNet.
- REST-like but over SWIFT-proprietary protocols.
- Supports FIN, InterAct, FileAct, and PKI.

C. SWIFT Alliance Messaging Hub (AMH)

- Large-scale message routing.
- Used by central banks, RTGS operators, and big Tier-1 banks.
- Plug-in engine + message transformation layer.

D. SWIFTNet InterAct / FileAct Clients

Used for:

- ISO 20022 MX
- SEPA
- CBPR+
- RTGS (e.g., TARGET2, FedNow future ISO)

✅ 2. FIX Engines (Market & Trading Financial Messages)

For securities, FX, crypto, trading, settlement:

Production FIX Engines

- **QuickFIX/J** (Java)
- **QuickFIX/C++**
- **QuickFIX/n**
- **FIX Antenna** (B2Bits)
- **OnixS FIX Engine**
- **CameronFIX**
- **Bloomberg BPIPE FIX Gateways**

These run over:

- Pure TCP
- TCP+TLS
- Raw TLS
- FIX-over-WebSocket (rare)

✅ 3. Real-time Payment Network Clients (Domestic RTP, RTGS)

A. TARGET2 / T2 / TIPS gateways (EU)

Vendors:

- Sopra Banking
- SIA / Nexi
- Perago RTGS frameworks
- SWIFT AMH modules

B. CHAPS / RTGS (UK)

- Bottomline RTS
- Finastra Fusion
- SWIFT RTGS modules (AMH/SAA)

C. Fedwire / FedNow (US)

- Volante Fedwire Gateway
- Finastra Quantum
- FIS / Clear2Pay payment hub

D. SEPA Direct/Instant

- PPI ICCS / SIANet
- EquensWorldline
- ACI RTPS
- Intix / Pelican

✅ 4. Host-to-Host Corporate Channels (Direct Bank-to-Corporate)

These are typical **IP-to-IP**, **server-to-server**, **non-SWIFT** secure channels.

A. EBICS (EU)

- XML payloads over TLS + signatures.
- Used in France, Germany, Austria.

B. SFTP/SSH with digital signatures

Used for:

- Bulk payments (XML)
- MT940/M940 statements
- Corporate treasury

C. AS2 / AS4 Gateways

Enterprise-grade, signature + encryption systems:

Vendors:

- Cleo Harmony / Cicero
- Axway B2Bi
- Seeburger BIS
- IBM Sterling B2Bi

These transport:

- ISO 20022 XML
- EDI
- files
- custom payloads

✅ 5. MQ-based Financial Messaging (Host to Host)

Major banks use **message queues** instead of HTTP.

A. IBM MQ over TLS

The most common **internal bank-to-bank** transport.

B. Apache Kafka (TLS + Avro/Protobuf)

Used in modern payment hubs.

C. TIBCO EMS / RV

Used by large trading desks, treasury and settlement systems.

✅ 6. Direct Raw TLS / Binary Payment Engines

These are rare but real for **direct IP-to-IP** financial connections.

Examples:

- **ISO 8583 Card Payment switches** (ACI, HPS PowerCARD, FIS)
- **Open Payment Framework (FIS)**
- **ACI Money Transfer System (MTS)**
- **RippleNet / Interledger connectors** (TLS binary frames)
- **Core banking middleware custom TLS sockets**

These transport:

- ISO 8583
- Custom bank XML
- ISO 20022
- MT-like payloads
- Binary TLV

✅ 7. Payment Hubs With Built-In Messaging Engines

These are "meta-gateways" that connect to SWIFT, SEPA, RTGS, CHAPS, Fedwire.

Major vendors:

- **Volante Designer / Payments Hub**
- **FIS** (multiple hub products)
- **Icon Solutions IPF**
- **Pelican AI Engine**
- **Finastra Global PAYplus**
- **Temenos Payments Hub**
- **TCS BaNCS Payments**
- **ACI Enterprise Payments Platform**

They support:

- MT & MX
- ISO 20022 validation
- TLS, MQ, SFTP, AS2, host-to-host
- Idempotency, rerouting, settlements

✅ 8. Other Command-line / terminal tools and libraries:

- openssl s_client
- gnutls-cli
- ncat –ssl
- curl --tlsv1.2 --http0.9
- socat openssl-connect
- stunnel

✅ 9. Programming languages

Any normal TCP+TLS library can send **financial messages** (ISO 20022, MT103, FIX, custom XML, etc.) without HTTP.

A. Python

B. Java

C. Go

Essentially **any language with TLS support can do this**.
